

FIBOQUADRATIC NUMBERS AND RITHMOMACHIA

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Abstract

This paper shows our discovery of the fiboquadratic numbers, quadratic Fibonacci numbers in rithmomachia. This board game consists in performing mathematical operations to attack an opponent through geometric numbered pieces. We will show that the numbers on the board pieces are the first terms of six sequences that we will define through a recursive pattern of superpartients. We then prove that this extension is the solution of a recurrence relation with initial conditions, and therefore, is unique. The fiboquadratic numbers we introduce in this work are not limited to the context of this board. In general, we can generate any fiboquadratic sequence with arbitrary initial conditions.

Mathematics Subject Classification: 97A20, 11B39, 65Q30.

Keywords: Rithmomachia, fiboquadratic numbers.

1 Introduction

Rithmomachia is a number board game created to provide a tool to support mathematical education in medieval universities and monastic schools. Its design used the concepts of Greek number theory found in *De Arithmetica* by Nicomachus (60-120 A.D.), which Boethius later translated (circa 480-524 A.D.). This paper shows that the numbers on the board are the first terms of six sequences of numbers that we will define throughout our work. We want to define the sequences, such that the original definitions of superparticulars (See 2.2) and superpartients (See 2.3), coming from Nicomachus and Boethius, be reached recursively by induction.

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The main achievement of our work is the discovery of a set of number sequences based on the quadratic combination of Fibonacci numbers, which we coin as *fibonacci quadratic numbers*. We generate the fibonacci quadratic numbers with the actual numbers we see on the board of rithmomachia. By extending, a recursive pattern, we keep the definitions of the numerical distribution of numbers. We will also prove that our fibonacci quadratic numbers are the solution of a linear, homogeneous recurrence relation of fourth-order with constant coefficients and initial conditions given by the first numbers on the board. Therefore, we prove they are unique.

Some branches such as Combinatorics have studied fibonacci quadratic numbers. For example, the sequence A006498 from OEIS counts the number of compositions of n with 1, 3, and 4 (See [Oeis98]). Meanwhile, sequence A006499 counts the number of restricted circular combinations (See [Oeis99]). In other words, we provide an alternative approach to concepts of modern mathematics by studying patterns previously given in a medieval board game. As a result, we generate more sequences than the two mentioned above with the fibonacci quadratic numbers. Interestingly, none of these numbers are in the OEIS.

1.1 Brief History of the Game

The oldest known version of rithmomachia originated from a letter from Asilo von Würzburg. Throughout the Middle Ages and the Renaissance, many scholars created several versions and manuals of the game. These manuals contained different playing styles with variations from the pieces' movements to the board's size. In different manuals, circles could move diagonally, in others vertically and horizontally. The same happened with the triangles and squares. Another disagreement was the number of spaces a piece was allowed to move. In some versions, circles could move one space, triangles two spaces, and squares three spaces, but that differed depending on the author's rules. Among the leading authors, we found Claude de Bossière, Ralph Lever and William Fulke, Benedetto Varchi, Carlo Strozzi, and Francesco Barozzi, etc (See [Moy01], pp. 1). These authors made significant contributions to the spread of the game in their respective eras. The authors mainly came from France (See [Moy01], pp. 88), England (See [Moy01], pp. 95), and Italy (See [Moy01], pp. 109) because the medieval society was decentralized. Even though the goal of this paper is not to present a particular version of the rules, we show in Figure 1.1 one of the most common boards today. Our results do not follow any rule of rithmomachia or board size but the distribution of numbers on the pieces, which is the same.

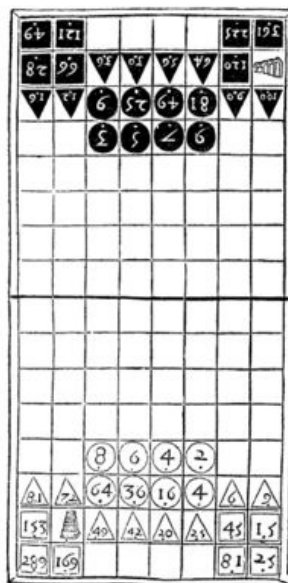


Figure 1: One of the original boards of rithmomachia presented by Claude de Boissière (See [Bos56], pp. 18).

1.2 Quick Summary on How to Play

As we stated before, there are many variations of the rules. In Ralph Lever and William Fulke's manual, circles move one space diagonally, triangles move two spaces vertically or horizontally, squares move three spaces vertically or horizontally, and the pyramids move like all the other pieces combined. These are considered regular moves. In addition, there's an irregular move that adds an extra space to the regular move, but in the perpendicular direction.

One makes a mathematical operation to capture a piece with either the values on the pieces or the values and spaces that separate them. There are many different ways to capture a piece. There's capture by encounter when two pieces of the same value meet from a regular move away. By ambush, when two attacking pieces meet the third piece, the values hold any of the four basic arithmetic operations. By assault, when through multiplication or division, the values and spaces allow you to capture the opponent's piece. By power and root, which is assault, but with the power or root operations. By progression, which looks like an ambush, but with the involvement of arithmetic, geometric, or harmonic progression, and finally, by surrounding, which is when you block a piece's ability to move to prevent it from attacking. Multiple captures with one single movement are allowed. Pieces are bicolored. At any point of the game, any player can turn one of the captured pieces in their own color and return it to the board. However, this is considered a move, and the player loses the points of the returned piece.

To win, you must fulfill the requirements of either a minor or major victory. We have five minor victories: De Bonis, by collecting a previously agreed upon minimum point value. De Corpore, by collecting a certain number of pieces. De Lite, by reaching a set score within a specific digit limit of the captured pieces. De Honore, by points and the maximum number of pieces captured, and De Honore Litique, total score, pieces, and digits. On the other hand, major victories include Magna, when the player places three pieces on the adversary's field in any of the three progressions (arithmetic, geometric, or harmonic). Superior, when four pieces are such that three are in any progression and three of them are in any other progression, and Excellentissima happens when from four pieces it is possible to form the three progressions with three pieces (not the same three pieces for the three progressions). Lever and Fulke's manual appears as an appendix of [Moy01]. We refer the reader to it for further reading about the rules.

2 Rithmomachia's connection with Greek number theory

Rithmomachia¹ is a medieval board game whose origin links to the teachings of arithmetic, found in *De Arithmetica* by Nicomachus translated by Boethius (See [Sch68] pp. 615). More than just a book of arithmetic in the current sense, *De Arithmetica* classifies the elements of a set of natural numbers $\mathbb{N}$. Inside of these classifications (See [Ger26] of Gerasa, from chapter XVIII to chapter XX, pp. 214-220), three definitions will be essential to the description of rithmomachia and the goals we want to achieve in this work. The first one is well known to us:

Definition 2.1. *A number p is a **multiple** of q if there's a number n such that $p = nq$.*

Note that these last two definitions are less familiar to us:

Definition 2.2. *A number p is a **superparticular** of q if there's a number n such that $p = (1 + \frac{1}{n})q$.*

Definition 2.3. *A number p is a **superpartient** of q if there are coprime numbers k and n such that $p = (1 + \frac{k}{n})q$ and $2 \leq k \leq n - 1$.*

These definitions require that q be a multiple of n .

The board starts with the set $\{2, 3, 4, 5, 6, 7, 8, 9\}$, and each piece is assigned a number² following the scheme in Table 1, described as follows:

1. Evens and odds are separated. (In the game, the evens will be on the white pieces and the odds on the black pieces. We keep this convention in our description. Even though it doesn't matter in terms of the arithmetic we're looking at.)
2. Each row modifies the numbers of the previous row as shown in the left column.

¹For a detailed history of the game, we refer the reader to Moyer [Moy01]. For an educational approach to Rithmomachia, see Smith and Eaton [SE11].

²Also, it is assigned a shape. However, this is not relevant here.

	White Pieces				Black Pieces			
$p = n$	2	4	6	8	3	5	7	9
$q = np$	4	16	36	64	9	25	49	81
$r = (1 + \frac{1}{n})q$	6	20	42	72	12	30	56	90
$s = (1 + \frac{1}{n})r$	9	25	49	81	16	36	64	100
$t = (1 + \frac{n}{n+1})s$	15	45	91	153	28	66	120	190
$u = (1 + \frac{n}{n+1})t$	25	81	169	289	49	121	225	361

Table 1: Number distribution of rithmomachia.

3. The second row is a multiple of the first row.
4. The third row is superparticular of the second row, and the fourth row is superparticular of the third row.
5. The fifth row is a superpartient of the fourth row, and the sixth row is a superpartient of the fifth row.

An easy algebraic calculation eliminates the fractions on the first column to get a polynomial expression, as shown in Table 2.

	White pieces				Black pieces			
$p = n$	2	4	6	8	3	5	7	9
$q = n^2$	4	16	36	64	9	25	49	81
$r = n(n+1)$	6	20	42	72	12	30	56	90
$s = (n+1)^2$	9	25	49	81	16	36	64	100
$t = (n+1)(2n+1)$	15	45	91	153	28	66	120	190
$u = (2n+1)^2$	25	81	169	289	49	121	225	361

Table 2: Polynomial distribution of rithmomachia.

We call the above tables *rithmomachia's board*. From now on, we won't separate the evens from the odds, as shown in Table 3.

3 Sequences created from rithmomachia's Numbers

A review of the formulas to the right of the board show us the beginning of a pattern that we can follow better when adding another row to the front of Table 3. Let us take

1	1	1	1	1	1	1	1	1
n	2	3	4	5	6	7	8	9
n^2	4	9	16	25	36	49	64	81
$n(n+1)$	6	12	20	30	42	56	72	90
$(n+1)^2$	9	16	25	36	49	64	81	100
$(n+1)(2n+1)$	15	28	45	66	91	120	153	190
$(2n+1)^2$	25	49	81	121	169	225	289	361

Table 3: Order distribution of rithmomachia.

the algebraic expressions as pairs:

$$\begin{array}{cc}
 1 & n \\
 n^2 & n(n+1) \\
 (n+1)^2 & (n+1)(2n+1) \\
 (2n+1)^2 & (2n+1)(\quad ? \quad)
 \end{array}$$

Now, let's take a look at the pieces of rithmomachia and how they place on the board (See [Moy01], pp. 11) as seen in the figure 2.

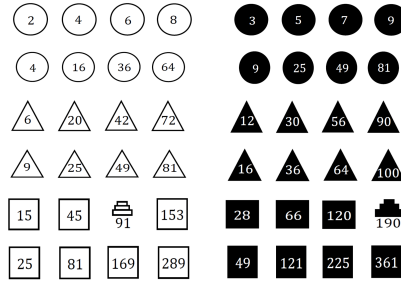


Figure 2: The pieces of rithmomachia by shapes.

We then notice from the diagram above that the pieces of rithmomachia hold for the following pattern in the next diagram of the figure 3. As we order the shapes (circles, triangles, and squares) in this way, we see the Fibonacci recurrence relation on the board.

Besides, this pattern reminds us of the pattern that generates the Fibonacci numbers.

The above prompts us to then define sequences with the numbers from the board as the initial conditions. The sequences will have Fibonacci numbers as coefficients of their terms.

$$\begin{array}{c}
\textcircled{2} + \textcircled{4} = \triangle 6 \\
\bullet 3 + \bullet 9 = \blacktriangle 12 \\
\triangle 6 + \triangle 9 = \square 15 \\
\blacktriangle 12 + \blacktriangle 16 = \blacksquare 28
\end{array}$$

Figure 3: The Fibonacci properties within the pieces.

This pattern then continues with all of the pieces of rithmomachia's board, starting with the two rows of circles adding to become the first row of triangles and ending with the two rows of triangles adding to become the first row of squares.

To the left, we see affine maps in n raised to the square. To the right, the affine maps repeat, but now multiplied by a new affine map, it will be the one to be squared in the next row. In this case, we can be more explicit with the table:

$$\begin{array}{cc}
(0n+1)^2 & (0n+1)(1n+0) \\
(1n+0)^2 & (1n+0)(n+1) \\
(1n+1)^2 & (1n+1)(2n+1) \\
(2n+1)^2 & (2n+1)(?n+?)
\end{array}$$

to see that the coefficients of the affine maps come from the *extended Fibonacci sequence*:

$$\dots, 5, -3, 2-1, 1, 0, 1, 1, 2, 3, 5, \dots$$

defined by:

$$F_0 = 0, \quad F_1 = 1, \quad F_{m+1} = F_m + F_{m-1}, \quad F_{-m} = (-1)^{m-1}F_m, \quad \text{and } m \geq 1$$

such that, if we enumerate the rows from $m = 0$, the collection of pairs above describe:

$$(F_m n + F_{m-1})^2 \quad (F_m n + F_{m-1})(F_{m+1} n + F_m), \quad m = 0, 1, 2, \dots \quad (1)$$

which allows us to answer the question and complete the subsequent rows:

$$\begin{array}{cc}
1 & n \\
n^2 & n(n+1) \\
(n+1)^2 & (n+1)(2n+1) \\
(2n+1)^2 & (2n+1)(3n+2) \\
(3n+2)^2 & (3n+2)(5n+3) \\
(5n+3)^2 & (5n+3)(8n+5) \\
\vdots & \vdots
\end{array}$$

$(a_n)_m$	$(a_2)_m$	$(a_3)_m$	$(a_4)_m$	$(a_5)_m$	$(a_6)_m$	$(a_7)_m$	$(a_8)_m$	$(a_9)_m$
1	1	1	1	1	1	1	1	1
n	2	3	4	5	6	7	8	9
n^2	4	9	16	25	36	49	64	81
$n(n+1)$	6	12	20	30	42	56	72	90
$(n+1)^2$	9	16	25	36	49	64	81	100
$(n+1)(2n+1)$	15	28	45	66	91	120	153	190
$(2n+1)^2$	25	49	81	121	169	225	289	361
$(2n+1)(3n+2)$	40	77	126	187	260	345	442	551
$(3n+2)^2$	64	121	196	289	400	529	676	841
$(3n+2)(5n+3)$	104	198	322	476	660	874	1118	1392
$(5n+3)^2$	169	324	529	784	1089	1444	1849	2304
$(5n+3)(8n+5)$	273	522	851	1260	1749	2318	2967	3696
$(8n+5)^2$	441	841	1369	2025	2809	3721	4761	5929
$(8n+5)(13n+8)$	714	1363	2220	3285	4558	6039	7728	9625
$(13n+8)^2$	1156	2209	3600	5329	7396	9801	12544	15625
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

Table 4: Sequences coming from rithmomachia.

This work results in the vertical extension of Table 3, now in the indicated form of Table 4.

A question that makes sense is: does this distribution of rows respect the original pattern of rithmomachia based on the definitions coming from Boethius? The answer is yes, with the understanding that from one row to the next, the most recent elements are superpartients of the elements immediately above, obtained by multiplication by the factor.

$$1 + \frac{F_{m-1}n + F_{m-2}}{F_m n + F_{m-1}} = \frac{(F_m + F_{m-1})n + (F_{m-1} + F_{m-2})}{F_m n + F_{m-1}} = \frac{F_{m+1}n + F_m}{F_m n + F_{m-1}}$$

In fact, regarding the expressions of (1), we have

$$(F_m n + F_{m-1})^2 \frac{F_{m+1}n + F_m}{F_m n + F_{m-1}} = (F_m n + F_{m-1})(F_{m+1}n + F_m)$$

and

$$(F_m n + F_{m-1})(F_{m+1}n + F_m) \frac{F_{m+1}n + F_m}{F_m n + F_{m-1}} = (F_{m+1}n + F_m)^2$$

which verifies the repeating of the pattern.

4 Fiboquadratic sequences

The quadratic formulas to the left of Table 4.4 define, to each value of n , a number sequence; we have given a name to the set of these sequences, in the following definition:

Definition 4.1. *For each n , the **fiboquadratic sequence of index n** , is defined by*

$$(a_n)_m = \begin{cases} (F_p n + F_{p-1})^2, & \text{if } m = 2p \\ (F_p n + F_{p-1})(F_{p+1} n + F_p), & \text{if } m = 2p + 1 \end{cases} \quad (2)$$

In equation (2), the index m can be any integer, though we will preferably handle it in a set of positive integers. Next, we will see some of the properties of fiboquadratic sequences to make this goal. We use the following notation:

$$\alpha = \frac{1 + \sqrt{5}}{2}, \quad \beta = \frac{1 - \sqrt{5}}{2}$$

For the roots of the golden equation, $x^2 = x + 1$ and α is the Golden Ratio. Our first theorem shows how to get the Golden Ratio from the fiboquadratic numbers.

Theorem 4.2. *If m is positive, the sequence of the ratios of consecutive terms of the fiboquadratic sequence converges to α for any index n .*

Proof: We have two cases: m is even, or m is odd. Let's see each.

a) $m = 2p$; $m + 1 = 2p + 1$. Then,

$$\begin{aligned} \frac{(a_n)_{m+1}}{(a_n)_m} &= \frac{(F_p n + F_{p-1})(F_{p+1} n + F_p)}{(F_p n + F_{p-1})^2} \\ &= \frac{F_{p+1} n + F_p}{F_p n + F_{p-1}} \\ &= \frac{\frac{F_{p+1}}{F_p} n + 1}{n + \frac{F_{p-1}}{F_p}} \end{aligned}$$

b) $m = 2p + 1$; $m + 1 = 2(p + 1)$. Then,

$$\begin{aligned} \frac{(a_n)_{m+1}}{(a_n)_m} &= \frac{(F_{p+1} n + F_p)^2}{(F_p n + F_{p-1})(F_{p+1} n + F_p)} \\ &= \frac{F_{p+1} n + F_p}{F_p n + F_{p-1}} \\ &= \frac{\frac{F_{p+1}}{F_p} n + 1}{n + \frac{F_{p-1}}{F_p}} \end{aligned}$$

The same in both cases. Also, when $p \rightarrow \infty$.

$$\frac{\frac{F_{p+1}}{F_p}n + 1}{n + \frac{F_{p-1}}{F_p}} \longrightarrow \frac{\alpha n + 1}{n + \frac{1}{\alpha}} = \alpha$$

□

We know that general sequences G_m of Fibonacci numbers are defined by

$$G_0 = a, \quad G_1 = b, \quad G_{m+1} = G_m + G_{m-1}, \quad m \geq 1$$

where a and b are any integer, both not zero, so that $G_m = F_m b + F_{m-1} a$. In the case of fiboquadratic sequences $(a_n)_m$, it's observed that factors of the form $G_p = F_p n + F_{p-1}$ correspond to a general Fibonacci sequence, with $G_0 = 1$ and $G_1 = n$. These considerations then lead us to a general concept of a fiboquadratic sequence.

Definition 4.3. *If $\{G_m\}_{m=0}^\infty$ is a general Fibonacci sequence, then the **general fiboquadratic sequence** generated from $\{G_m\}$ is the sequence*

$$a_m = \begin{cases} G_p^2, & \text{if } m = 2p \\ G_p G_{p+1}, & \text{if } m = 2p + 1 \end{cases} \quad (3)$$

Now, we want to prove the following property of a fiboquadratic sequence.

Theorem 4.4. *A general fiboquadratic sequence $\{a_m\}_{m=0}^\infty$ as in (3) satisfies the next recurrence relation*

$$a_m = a_{m-1} + a_{m-3} + a_{m-4}, \quad \text{with} \quad a_0 = a^2, \quad a_1 = ab, \quad a_2 = b^2, \quad \text{and} \quad a_3 = (a+b)b \quad (4)$$

Proof: Suppose m is even, then

$$a_m = G_p^2, \quad a_{m-1} = G_{p-1}G_p, \quad a_{m-3} = G_{p-2}G_{p-1} \quad \text{and} \quad a_{m-4} = G_{p-2}^2$$

So,

$$\begin{aligned} a_{m-1} + a_{m-3} + a_{m-4} &= G_{p-1}G_p + G_{p-2}G_{p-1} + G_{p-2}^2 \\ &= G_{p-1}G_p + (G_{p-1} + G_{p-2})G_{p-2} \\ &= G_{p-1}G_p + G_p G_{p-2} \\ &= G_p(G_{p-1} + G_{p-2}) \\ &= G_p^2 \\ &= a_m \end{aligned}$$

On the other hand, if m is odd, then,

$$a_m = G_p G_{p+1}, \quad a_{m-1} = G_p^2, \quad a_{m-3} = G_{p-1}^2 \quad \text{and} \quad a_{m-4} = G_{p-2}G_{p-1}$$

Therefore,

$$\begin{aligned}
 a_{m-1} + a_{m-3} + a_{m-4} &= G_p^2 + G_{p-1}^2 + G_{p-2}G_{p-1} \\
 &= G_p^2 + G_{p-1}(G_{p-1} + G_{p-2}) \\
 &= G_p^2 + G_{p-1}G_p \\
 &= G_p(G_p + G_{p-1}) \\
 &= G_pG_{p+1} \\
 &= a_m
 \end{aligned}$$

The initial conditions $a_0 = G_0^2 = a^2$, $a_1 = G_0G_1 = ab$, $a_2 = G_1^2 = b^2$, and $a_3 = G_1G_2 = (a+b)b$ are verified in (4). $\square$

Since the fiboquadratic numbers are the solution to a recurrence relation, the original pattern from the board, according to the definitions from Greek number theory, is maintained. We can conclude from Theorem 4.4 that Table 4 continues the pattern given by the equations 4.1. Each sequence of Table 4 contains some of the numbers of rithm-machia's board as the initial terms of a family of six sequences of numbers, according to ([Mil60] Miller, Theorem 1 pp. 127). These sequences are unique if we take the numbers on the board as the initial conditions.

Finally, we want to show the solutions to the fiboquadratic sequences $\{(a_2)_m\}_{m=0}^\infty$ and $\{(a_3)_m\}_{m=0}^\infty$. We can do this by solving (4) through its' characteristic polynomial $r^4 - r^3 - r - 1 = 0$, which factors as $(r - \alpha)(r - \beta)(r - i)(r + i) = 0$. Therefore, a general solution to (4) is

$$a_m = C_1\alpha^m + C_2\beta^m + C_3\cos\frac{m\pi}{2} + C_4\sin\frac{m\pi}{2} \quad (5)$$

The fiboquadratic sequence $\{(a_2)_m\}_{m=0}^\infty$ is (5) with initial conditions $a_0 = 1$, $a_1 = 2$, $a_2 = 4$, and $a_3 = 6$. After solving the linear system, we find that

$$(a_2)_m = \frac{1}{5} \left[\alpha^{m+4} + \beta^{m+4} - 2\cos\frac{m\pi}{2} - \sin\frac{m\pi}{2} \right]$$

Recall, the *Lucas numbers* are defined by:

$$L_0 = 2, \quad L_1 = 1, \quad L_{m+1} = L_{m-1} + L_m, \quad \text{and} \quad m \geq 1$$

As $L_m = \alpha^m + \beta^m$, the linear expression of $\{(a_2)_m\}_{m=0}^\infty$ is

$$(a_2)_m = \frac{1}{5} \left[L_{m+4} - 2\cos\frac{m\pi}{2} - \sin\frac{m\pi}{2} \right] \quad (6)$$

We notice that (6) is the truncated version of the sequence A006498. In the same way, the fiboquadratic sequence $\{(a_3)_m\}_{m=0}^\infty$ is (5) with initial conditions $a_0 = 1$, $a_1 = 3$, $a_2 = 9$, and $a_3 = 12$. In other words,

$$(a_3)_m = L_{m+2} - 2\cos\frac{m\pi}{2} - \sin\frac{m\pi}{2} \quad (7)$$

The equation (7) is the complete sequence A006499. Both sequences A006498 and A006499 are in the OEIS (See [Oeis98] and [Oeis99] for further details). An expression

to the complete sequence A006498 is the fiboquadratic sequence $\{(a_1)_m\}_{m=0}^{\infty}$ which we do not consider in this paper. However, it's just a shift of (6). That is,

$$(a_1)_m = \frac{1}{5} \left[L_{m+2} + 2 \cos \frac{m\pi}{2} + \sin \frac{m\pi}{2} \right] \quad (8)$$

5 Conclusion

The numbers in Table 4 are sequences that continue the pattern of superpartients recursively. The initial numbers on the board can generate other sequences. However, the fiboquadratic sequences that we studied in this work are consistent with the historical ideas that created rithmomachia.

It is surprising how medieval scholars anticipated concepts of modern mathematics just with elementary ideas of Greek number theory in the design of a board game to support the study of the Quadrivium.

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